

## 1. Project Title - SRSly Fit

## 2. Project Summary

It can be a struggle to improve your health if you are unsure of where to start or if you are too busy to dedicate much time to it. We propose a workout companion website that will make the process easier for beginners and veterans alike to create workouts, keep track of progress, and learn about new exercises. The user will be able to manually and automatically generate workouts for specific muscle groups with the number of sets, reps, and weight for previously completed exercises automatically carried over for ease of use.

## 3. Description

The website will automatically generate workouts for the user based on a series of factors (explained in more detail below). The website can be used to educate the user more about how to workout through viewing the pages set aside for each exercise. The exercise pages will contain descriptions/instructions for the exercises and embedded youtube videos.

Additionally, the site will serve as a “workout logger” which will allow the user to update their workout progress while working out. They will be able to update their preference for each exercise, swap out exercises, mark when exercises are completed, and update number of sets and reps and amount of weight lifted. The “workout logger” feature will make it easier to manually create or automatically generate workouts in the future because previously completed exercises will be prepopulated with the last completed number of sets, reps, and amount of weight lifted for user convenience.

The user will also have the ability to filter exercises by the equipment they have available, their preference towards exercises, and the muscle group of the exercises when manually creating or generating workouts. Furthermore the workout companion will suggest which muscle groups the user might be neglecting/needing to focus on more based on previous volume/intensity in past workouts.

For a beginner lifter, the website/app would focus on high-level compound lifts and then from there once all major muscle groups have been hit, it could suggest supplementary lifts that will target auxiliary muscle groups that might be normally neglected.

## 4. Creative Component(s)

We propose 3 creative components. The first is a more advanced visualization component that allows the user to click the muscles on a human to select which muscle(s) they would like to filter exercise results with. This feature will be used when manually and automatically creating workouts. By visually showing the user which muscles they would like to workout, beginner lifters won't have to know much about anatomy in order to know which muscle groups they would like to workout. In the event we are unable to make the visualization work properly our backup plan is to show a list of muscle groups to filter with.

The second component we propose is one that allows the user to automatically generate custom workouts based on a number of factors such as recently completed exercises, user preference of exercises, workout goals, and available workout equipment. Before generating the workout, the user will be able to specify the number of exercises and the muscle groups for the workout. The user's preference of the exercises and the equipment the user has available to them will already be stored based on previous feedback given by the user. This component will work by iteratively filtering out certain exercises that don't meet certain criteria before adding a random exercise from the final filtered list to the workout.

We would first filter the exercise options by the user's specified muscle group(s). Next, we will filter out the exercises that the user has "disliked". Finally, we would count the number of times the user has performed each of the remaining exercises and pick a random exercise that has been performed less than the maximum number of times (unless all of the counts are equal). In doing so, the user would be exposed to new exercises with every newly generated workout. To make this process even easier for the user, we may additionally provide suggested templates for the user to use with certain parameters already chosen to automatically generate workouts from.

It can be difficult to know how to perform an exercise without ever having seen it performed before. Therefore, the website will have pages dedicated towards informing the individual about the exercise they have selected. For our third component, we propose another feature which will use the Youtube API to embed an instructional workout video on each exercise page so the user can watch the video to understand how to perform the exercise. We will use the Youtube API to search for the most relevant video and embed the closest match(es) on the page.

## 5. Usefulness

From our initial research in workout websites and applications, we came across the Strong app, Hevy app, and MuscleWiki website which stood out to us for various reasons. Our website will take some inspiration from these sources while additionally building on with new features.

The Strong app is a helpful tool which allows you to plan and track your workout progress as you're working out. It is a convenient tool to document your workout progress and makes it easier to return to old exercises since it automatically populates the exercises with the number of sets, reps, and amount of weight you previously lifted. While it supports "templates" that allow you to plan workouts in advance, it is not very friendly for beginners who don't know how to create workouts or for busy individuals who don't have the time to do so. The app does not generate the workouts for you (outside of a few basic pre-generated workout examples that are provided). It also requires the user to understand the basics of working out in order to manually create workouts in the app.

The Hevy app provides some anatomical visualization of muscle groups that have been activated in the workout, but doesn't give a good representation of the intensity. We can

supplement this info by scaling the color of each muscle group to represent the volume/load of exercises to highlight how much focus was placed on a specific muscle group within a certain week's workouts. Furthermore, we can add the anatomical visualization along with the workout generator/selector to actively show the user what muscle will be hit in their active sessions vs showing the activated muscles by the end of the session. It also has workout logging capabilities, but does not include suggested exercise weights to guide a user into progressive overload.

The MuscleWiki website provides support for learning about new exercises with a user-friendly search functionality that retrieves the most relevant exercise pages. These pages have embedded youtube videos and instructions that inform the user on how to perform the exercise. Additionally the website has a new workout generator tool that allows the user to automatically generate workouts for specific muscle groups. Unfortunately, it only generates workouts with 3 exercises (this cannot be modified), it does not allow you to choose multiple muscle groups when generating the workout (so exercises which tackle multiple muscle groups aren't necessarily prioritized if you wish to also hit those muscles in the same workout), and it does not allow you to track your lifting progress over time (so you will have to always guess at the amount of weight you should lift for their suggested amount of sets and reps).

We hope to combine the best of these features into a single website. From a high level, the features include: manual and automatic workout generation, workout progress logging, exercise searching, exercise pages with instructional information, user preferences towards workouts, availability of equipment, and search and filtering functionality.

More specifically, users will be able to add exercises and equipment to the database. Functionality will be added so users can rate their preference for each exercise as "like", "dislike", or "Null". They will additionally be able to filter based on their preference and the equipment they have available when searching for exercises and automatically generating workouts. They will be able to update their profile with the equipment they have access to. A page will be dedicated towards each exercise complete with embedded youtube videos and a description/instructions so the user can inform themselves on how to perform it. When searching for exercises and creating workouts, users will be able to filter and search for exercises based on their type. They will be able to automatically and manually create workouts for muscle groups. As they are working out, they will be able to log their workout while it is in progress by adding/removing exercises from the workout, add/remove/update reps and sets for each exercise, reorder the exercises in the workout, and check mark the entire workout, individual exercise(s) and/or individual sets as completed. Inside of each workout, users will be able to "reroll" the exercise that is in the workout to automatically swap it out with a random exercise that meets the same filters originally specified. Users will be able to save previously completed workouts as workout templates that they can use to build future workouts off of. Assuming we incorporate a calendar into the website, users will be able to schedule and reschedule their workouts.

## 6. Realness / Datasets

We have found 3 data sources to potentially use for the project. The Gym Exercise Dataset (megaGymDataset.csv) from Kaggle contains 2,918 rows of data on gym exercises in a csv file. It contains the following columns: “Title”, “Desc” (Description), “Type” (e.g., Cardio vs Strength), “BodyPart” (the body part that is targeted), “Equipment”, “Level”, “Rating”, and “RatingDesc” (rating description). The dataset notably associates each exercise with a description, the body part targeted, and the equipment required.

<https://www.kaggle.com/datasets/niharika41298/gym-exercise-data>

API Ninjas provides an Exercises API which is a dataset of thousands of gym exercises returned in a JSON format. It contains the following columns: “Name”, “type”, “muscle”, “equipment”, “difficulty”, and “instructions”. The dataset also contains fewer columns than the previously mentioned dataset but all of the columns it does contain are in common with the previously mentioned dataset with some renaming.

<https://api-ninjas.com/api/exercises>

Free Exercise DB is an Open Public Domain Exercise Dataset which contains over 800 exercises that are returned in a JSON format. It contains the following columns: “id”, “name”, “force”, “level”, “mechanic”, “equipment”, “primaryMuscles”, “secondaryMuscles”, “instructions”, “category”, “images”. The dataset contains some columns in common with both previously mentioned datasets with some renaming with the addition of multiple muscle groups mentioned, images, and force and isolation data that can be additionally used for search filtering.

<https://github.com/yuhonas/free-exercise-db>

A notable mention that we may be able to use for our project is the Gym Members Exercise Dataset that is found on Kaggle. It is a csv file that contains columns for “Age”, “Gender”, “Weight”, “Height”, “Max\_BPM” (maximum heart rate beats per minute (BPM)), “Avg\_BPM (average BPM)”, “Resting\_BPM”, “Session\_Duration”, “Calories\_Burned”, “Workout\_Type”, “Fat\_Percentage”, “Water\_Intake”, “Workout\_Frequency”, “Experience\_Level”, “BMI” (Body Mass Index). We could potentially use this dataset to provide calorie expenditure estimates provided we have the age of the user and duration of their workout, assume a linear association between duration and calories spent, and assume that each exercise expends a similar amount of calories. We also discussed potentially incorporating the Apple HealthKit API and/or Fitbit API for calorie estimation. This can also be estimated without any API or dataset by using an equation:  $(\text{MET} \times \text{body weight(KG)} \times 3.5)/200 \approx \text{calories burnt per minute}$ . We know the value of MET for weight lifting is between 3 and 5, with the specific value being based on how vigorous the workout is.

<https://www.kaggle.com/datasets/valakhorasani/gym-members-exercise-dataset>

## 7. Detailed Description of Functionality

1. **A low-fidelity UI mockup:** What do you imagine your final application's interface might look like? A PowerPoint slide or a pencil sketch on a piece of paper works!

We will have basic account creation and sign in pages when initially accessing the website. Users can create accounts using the Account Creation page. They can sign into the website with the Sign In page. A User Preferences page will also exist which will allow the user to update what equipment they have access to. A Workout History page will be used to either show a log of previously completed workouts or graphs of relevant workout data so users can track their progress.

With the below Workout Page, users will be able to update the log with the status of their workout in between exercises. They will have the ability to “like”, “dislike”, “remove like/dislike” from each of the listed exercises, modify the values for the weight and repetitions, and add or remove sets. They will additionally be able to add and remove exercises. Once a workout is marked as completed, the workout, exercises, and all values will be stored in the database so we can potentially show historical metrics/graphs later for the user to see their workout progress over time.

Navigation Bar

Start Workout | History | Exercise Search | Profile

Workout Name  
Date

Exercise 1 Name

| Set | Previous | lbs | Reps |
|-----|----------|-----|------|
| 1   | 35 x 18  | 35  | 18   |
| 2   | 35 x 20  | 35  | 20   |
| 3   | 40 x 25  | 40  | 25   |

Add Set

Exercise 2 Name

| Set | Previous | lbs | Reps |
|-----|----------|-----|------|
| 1   | 35 x 18  | 35  | 18   |
| 2   | 35 x 20  | 35  | 20   |

Add Set

Dislike

Like

Exercise has been Disliked

Entire Exercise Marked as completed

Exercise has been Liked

Marked as completed

Workout Page

With the below Workout Creation Page, Users can click the Add Exercises button to open up a search window/page that is used to add more exercises to the workout. If a workout doesn't exist, the user will have the ability to click a button to add a new exercise to the database. Users can click the Add Set button to add additional sets to exercises and can delete sets that have already been added. For each set, the user can directly modify the Lbs and Reps fields to change the weight for the exercise and the amount of repetitions, respectively. A Save Workout button at the bottom of the screen will be used to save the workout to the database.

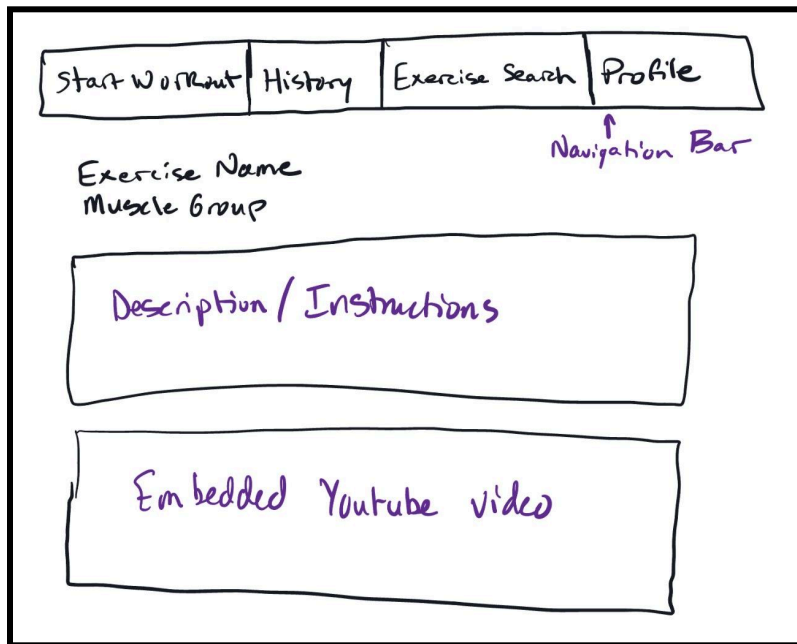
| Set | Previous | lbs | Reps |
|-----|----------|-----|------|
| 1   | 35 x 18  | 35  | 18   |
| 2   | 35 x 20  | 35  | 20   |
| 3   | 40 x 25  | 40  | 25   |

**Workout Creation Page**

With the below Exercise Search Page, Users will be able to search the database with keywords and filters to pull up a list of relevant exercises. They will be able to filter the results based on their preference of exercises, by muscle group, and by equipment. Additional filters may be added in the future.

**Exercise Search Page**

Users will be able to learn more about selected exercises with their individual Exercise Pages. These pages will show the name of the exercise, muscle group(s), description/instructions about the exercise, and embedded youtube video(s) to educate the user about the exercise. A “like/dislike” button will be added somewhere on the page which will let users update their preference for the exercise. We may additionally add a button that will allow the user to more conveniently add the exercise to pre-existing workouts (so that they don’t have to navigate to the workout and re-search for the exercise again).



**Exercise Page**

2. **Project work distribution:** Who will be responsible for each of the tasks or subtasks? Explain how backend systems will be distributed across members. Be as specific as possible as this could be part of the final peer evaluation metrics.

All members are expected to contribute across all roles, but the specified members will be the team leads for their respective sections of the project:

### **Member A: Team Lead & Integration Specialist - Lucas Eby**

**Primary Focus:** Project Coordination, Database Conceptual & Logical Design, Overall Backend Architecture, Deployment, and Final Report Compilation.

### **Member B: Database Architect & Data Steward - Serge Pod**

**Primary Focus:** Authentication & Core User Management, Frontend Integration, DDL, Data Ingestion, and Indexing.

### **Member C: Core API & Workout Management Logic - Rami Nasser**

**Primary Focus:** Implementing the primary CRUD operations and business logic for exercises and workouts, using raw SQL queries.

### **Member D: Advanced Features & Database Programs - Sean Kraemer**

**Primary Focus:** Implementing the "creative component" logic (especially workout generation), advanced SQL queries (Stored Procedures, Transactions, Triggers), and more complex data analysis.

For Stage 2: Database Design, we could divide up the data sources so that each source will have an expert:

1. Gym Exercise Dataset - Sean Kraemer
2. API Ninjas - Lucas Eby
3. Free Exercise DB - Serge Pod
4. Gym Members Exercise Dataset - Rami Nasser